

SUPPLEMENTARY SEMESTER EXAMINATIONS – JULY 2023

Program	: B.E :- Information Science and Engineering	Semester	: III
Course Name	: Object Oriented Programming with Java	Max. Marks	: 100
Course Code	: IS36	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT – I

- List and Explain the three types of variables we can create in Java with suitable examples. CO1 (08)
 - Write a Java program using for-each version to check the given number is prime or not. CO1 (06)
 - How Garbage Collection is done in Java? Explain how finalize() method works. CO1 (06)
- What is a constructor? Write a Java program to print length of a string entered by the user with the help of parameterized constructor. CO1 (08)
 - Explain the following control statements for labeled for loop with a code snippet: i) Break ii) Continue. CO1 (06)
 - Explain how instance variable hiding is done in Java with example. CO1 (06)

UNIT – II

- With the help of program, discuss overloading Constructor. CO2 (08)
 - Demonstrate the use of super with code snippets. CO2 (06)
 - Differentiate between static and final members of a class. CO2 (06)
- Create a class called account with the data members (Accno – integer, name- String, Phone_No - integer, balance_amt - float), and following methods :
 - getinput() to get input from the user
 - Deposit() method which takes the amount to be deposited in to his/her account and do the calculation.
 - Withdraw() method which gets the amount to be withdrawn from his/her account.
 - Print the appropriate results.
 - What are var-arg methods? Write Java code to demonstrate overloading of var-arg methods. CO2 (06)
 - With the help of program explain nested and inner class. CO2 (06)

UNIT – III

- What is Run Time Polymorphism? Explain Dynamic Method Dispatch with help of an example program. CO3 (08)
 - Create a Class Add with two private instance variable a and b, write a getter and setter method to access the variable. Create another class AddNumbers to access the method to add and print the result. CO3 (06)
 - What are the four access modifiers in Java? Tabulate their scope. CO3 (06)

6. a) Differentiate between an interface and an abstract class with suitable code snippets. CO3 (06)
- b) Write a Java Program that does the following related to Inheritance: CO3 (08)
- Create an abstract class called Vehicle which contains the 'year_of_manufacture' data member and two abstract methods 'getData()' and 'putData()' with a constructor.
 - Create two derived classes "TwoWheeler" and "FourWheeler" and implement the abstract methods. Make "FourWheeler" as final class.
- Create class 'MyTwoWheeler' which is a sub-class of "TwoWheeler" and demonstrate the use of super keyword to initialize data members of "MyTwoWheeler".
- c) What is an interface in Java? List the characteristics of interface variables and methods. CO3 (06)

UNIT – IV

7. a) Explain five keywords associated with Exceptions in Java with suitable examples. CO4 (08)
- b) Write a Program in Java to create a class to print factorial of a number and throw an exception if the entered value is not an integer value. Write a main method to take inputs from the user and call appropriate methods to print the output. CO4 (06)
- c) Demonstrate the working of nested try-catch block. CO4 (06)
8. a) List and explain any four exceptions that occur generally while code is running in Java. CO4 (08)
- b) Explain the use of throw and throws clauses in Java Exceptions through code snippets. CO4 (06)
- c) Demonstrate the creation of user-defined exceptions in Java and using the same in code. CO4 (06)

UNIT-V

9. a) Write a Java Program to add BookName to LinkedList object and enter names of 5 books. Sort the collection according to the alphabetical order. CO5 (08)
- b) Illustrate auto-boxing and auto-unboxing with suitable examples. CO5 (06)
- c) Explain any six methods in ArrayList class. CO5 (06)
10. a) List and explain any four Methods Defined by Collection interface. CO5 (08)
- b) What is wrapper class? How it is useful in Java? CO5 (06)
- c) Differentiate between Collection and arrays. CO5 (06)
